xv6-aarch64：ext2 为根时的函数调用链（register_filesystem / get_sb / fill_super 版）

左：调用树（缩进 = 调用层次）　中：文件:行号　右：日志里对应的那一行　橙色序号与挂载图 fig-k 的 ⓪–⑥ 对应

VFS / 挂载　xv6fs　ext2　块层 / 驱动　xjournal

A. 启动早期：登记文件系统类型（main，hart 0，还没有进程）		
• main()	main.c:36	
• binit() 建 bcache	main.c:85	
• iinit()	fs.c:279	
• fs_mountinit()	fs.c:915	
• register_filesystem(&xv6fs_type)	fs.c:918	register_filesystem: xv6fs
• do_initcalls()	main.c:93	
• [device] sdhost_probe → add_disk → msdos_partition	sdhost.c:484	mmcblk0: p1 p2 p3
• [late] rootdev_init()：root= 解析出 ROOT_DEV	rootdev.c:169	rootdev: ROOT_DEV = 5 (mmcblk0p3)
• [late 5] ext2init()：清空 ext2_instances[]	⓪ ext2.c:1114	
• register_filesystem(&ext2_type) 只登记，不读盘	vfs.c:1047	register_filesystem: ext2
• [late 6] vfsinit()	vfs.c:1194	
• register_filesystem(procfs / sysfs / fat32 / netfs)	vfs.c:1047	vfs: one mount table …; xv6fs ext2 procfs …
• userinit() → scheduler() → forkret()	proc.c:1188	
B. forkret → prepare_namespace()：rootfs、devtmpfs、挂 ext2 根、chroot		
• prepare_namespace()	do_mounts.c:405	
• ramfs_create(0, 7, "ram0")	do_mounts.c:84	
• get_fs_type("xv6fs") → xv6fs_fill_super(7, 0)	fs.c:132	
• blkdev_set_bsize(7,1024) → fs_readsuper(7) → fs_super_register	fs.c:88	fill_super: … dev=7
• devtmpfs_init() → ramfs_create(1, 8, "ram1") 同上，dev 8	do_mounts.c:354	devtmpfs: initialized on ram1
• kmknod("/dev/root", T_DEVICE, BLOCKDEV, 5)	do_mounts.c:426	
• mount_root("/dev/root", "/root")	do_mounts.c:390	
• vfs_get_sb("ext2", "/dev/root", VFS_MOUNT_ROOTLOG)	① vfs.c:1103	
• get_fs_type("ext2") 查 file_systems 链表	vfs.c:1060	
• ext2_type.get_sb = ext2_get_sb()	ext2.c:1087	
• lookup_bdev("/dev/root") → namei → (7,5)，minor = 5	② vfs.c:1071	lookup: success path=/dev/root -> (7,5)
• ext2_fill_super(5, flags)	③ ext2.c:1050	
• blkdev_get(5)：start、nsect、partno → ext2_instances[5]	blkdev.c:341	
• ext2_journal_init(fs) → xj_init() → recover()；binvalidate(5)	ext2.c:1026	xjournal: ready on mmcblk0p2+65536
• setup_ext2(fs, 1884160, …) → diskbytes(fs, 1024, sb)	ext2.c:958	ext2: p3 ready rw … block=4096
• e2_rsector()：先 xj_read(lba)，没有再 bread(5, 块号)	ext2.c:122	
• bread → bget → blk_rw → blkdev_get(5) → part_rw(+start)	bio.c:100	
• mmc_rw → sdsector(lba) 读 SD 卡扇区	sdhost.c:472	
• blkdev_set_bsize(5, 4096) bcache 改用 4 KiB 块	blkdev.c:362	
• ext2_register_super(5, 0)	④ ext2.c:2039	
• fs_super_register(5,"ext2",&ext2_iops,2,…,priv=&ext2_instances[5])	fs.c:51	register_super: fs=ext2 dev=5 … priv=0xffffff80…
• (vfs_get_sb 返回 dev = 5)	vfs.c:1103	get_sb: type=ext2 source=/dev/root -> dev=5
• klookup("/root") → (7,4)	⑤ do_mounts.c:115	lookup: success path=/root -> (7,4)
• fs_mount((7,4), 5, "ext2", "/dev/root", "/root") 填 mtab.m[0]	fs.c:1030	VFS: Mounted root (ext2 …) on device 5
• kchdir("/root")	⑥ do_mounts.c:132	
• namei → namex → follow_mount((7,4))	fs.c:962	follow_mount: cover=(7,4) -> fs=ext2 dev=5
• fs_dev_root(5) → iget_path(5, rootino=2)	fs.c:1011	iget_path: cache miss … dev=5 ino=2
• ilock → iop->read_inode = e2_read_inode	ext2.c:1823	ilock: … read_inode fs=ext2 dev=5 ino=2
• ext2_from_inode(ip)：supers[5].priv → readinode(fs, 2)	ext2.c:91	
• devtmpfs_mount("dev", "/root/dev") → fs_mount((5,76), 8)	do_mounts.c:368	mount: path=/root/dev cover=(5,76)
• fs_move_mount(from, (7,1), "/") mtab[0].mp: (7,4) → (7,1)	fs.c:1072	
• fs_set_root(p->cwd) "/" = (5,2)	fs.c:1108	VFS: root moved over rootfs, chroot done
C. /init 的 mount_fstab()：用户态 mount(2)		
• sys_mount()	sysfile.c:148	
• vfsmount(source, target, fstype, flags)	vfs.c:1157	
• vfs_get_sb(fstype, source, flags) → get_fs_type → t->get_sb	vfs.c:1103	
• procfs_get_sb / sysfs_get_sb / fat32_get_sb / netfs_get_sb	vfs.c:1115	get_sb: type=procfs source=proc -> dev=20
• fs_super_register(20 / 23 / 21 / 22, …) 伪文件系统，固定 dev	fs.c:51	
• ext2_get_sb("ext2")：source 不是设备节点 → 已有实例 dev 5	ext2.c:1087	get_sb: type=ext2 source=ext2 -> dev=5
• namei(target) → fs_mount(mp, dev, …)	fs.c:1030	vfs: mounted procfs at /proc read-only
• ext2：mount_of_dev(5) 已挂在 / → 返回 -1	fs.c:951	cannot mount ext2 at /mnt/ext2 (busy …)